Executive Summary – Diwali Sales Analysis

1. Project Overview

The objective of this project is to analyze Diwali sales data to understand customer purchasing behavior and identify the customer groups and product categories that contribute most to sales. I used Python, Pandas, Matplotlib, and Seaborn to clean the data, perform Exploratory Data Analysis (EDA), and create different charts to visualize customer demographics, purchasing patterns, and sales performance.

The original dataset contained **11,251 records and 15 columns**. After removing unnecessary columns and handling missing values, I retained **11,239 records across 13 columns**, preserving approximately **99.89% of the original records**.

2. Key Findings and Insights

Gender Analysis: Female customers accounted for 7,832 records (approximately **69.69%**), while male customers accounted for 3,407 records (approximately **30.31%**). This shows that female customers represented the larger share of buyers in the dataset.

Age Group Analysis: The charts indicate that customers aged **26–35 years** formed an important customer segment, particularly female customers. The age-group sales chart also helps compare the total purchase amount contributed by different age groups.

State-Wise Sales Analysis: Uttar Pradesh, Maharashtra, and Karnataka were identified in the notebook as the leading states in terms of orders and total sales. These states represent important markets for customer engagement and future sales campaigns.

Marital Status Analysis: The visualizations indicate that married female customers contributed substantially to purchasing activity and sales amount. This suggests that marital status, combined with gender, may be useful for understanding customer purchasing patterns.

Occupation Analysis: Customers working in the IT, healthcare, and aviation sectors were highlighted as important buyer groups in the notebook. Understanding these

customers' purchasing behavior can help businesses plan more relevant marketing campaigns.

Product Category Analysis: Food, clothing, and electronics were identified in the notebook as leading product categories in terms of sales activity. These categories deserve attention when planning product promotions, inventory, and festive offers.

Product-Level Analysis: I also created a chart showing the products with the highest number of orders. This helps identify popular products and provides a starting point for inventory planning and product-level promotions.

3. Business Value

This analysis helps transform raw sales data into useful business information. The charts make it easier to understand who is purchasing, where customers are located, which occupations are represented, and which product categories attract buyers. These insights can support better marketing decisions, inventory planning, customer targeting, and sales strategies during festive seasons.

My Recommendations

Based on my analysis, I recommend the following actions:

1. Focus on the largest customer segments.
Since female customers account for approximately 69.69% of the records, the business should understand their product preferences and shopping needs. Marketing should remain inclusive and be guided by actual customer behavior rather than assumptions.

2. Target customers aged 26–35.
This age group, particularly female customers, was highlighted in my analysis. The company can test targeted festive offers, product bundles, and personalized promotions for this segment.

3. Strengthen marketing in leading states.
Uttar Pradesh, Maharashtra, and Karnataka were identified as important markets. Businesses can focus on regional campaigns, local customer preferences, and timely stock availability in these states.

4. Create relevant campaigns for different customer groups.
The analysis highlights married women and customers working in IT, healthcare, and

aviation. Businesses can use these observations to develop customer segments and test tailored offers, while validating the results with further analysis.

5. Prioritize popular product categories.
Food, clothing, and electronics were highlighted as leading categories. Maintaining sufficient stock, offering suitable product combinations, and promoting these categories during festive periods may help support sales.

6. Use product-level insights for inventory planning.
The top-products chart can help identify frequently ordered items. Businesses should combine order counts with sales amounts, profit margins, and stock levels before deciding which products to prioritize.

7. Improve future analysis using percentages.
I recommend calculating each group's share of total orders and total sales, as well as the average purchase amount per customer. These metrics would make comparisons between genders, age groups, states, and occupations more meaningful.

8. Track results after campaigns.
Businesses should compare sales, order volume, and average purchase value before and after promotions. This would help determine whether marketing campaigns are actually improving performance.

Conclusion

This project helped me understand how data cleaning, exploratory data analysis, and visualization can reveal useful patterns in customer purchasing behavior. The findings highlight female customers, the 26–35 age group, important states, selected occupations, and popular product categories as areas worth further investigation.

By using these insights to improve marketing, product availability, and customer targeting, businesses can make better-informed decisions and plan more effectively for future festive sales periods.